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Sri Lanka Institute of Information Technology

Faculty of Computing

Matrices

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Matrices – definitions

- A matrix is a set of real or complex numbers (called elements) arranged in rows and columns to form a rectangular array.
- A matrix having m rows and n columns is called an $m \times n$ matrix.
- For example:

$$\begin{bmatrix} 2 & 0 & 1 \\ 3 & 0 & 0 \\ 5 & 1 & 1 \end{bmatrix}$$

is a 2×3 matrix.

Matrices – definitions

- Row matrix
  - A row matrix consists of a single row. For example:

$$\begin{bmatrix} 4 & 3 & 7 & 2 \end{bmatrix}$$

- Column matrix
  - A column matrix consists of a single column. For example:

$$\begin{bmatrix} 6 \\ 3 \\ 8 \end{bmatrix}$$

Matrices – definitions

- Double suffix notation
  - Each element of a matrix has its own address denoted by double suffices, the first indicating the row and the second indicating the column. For example, the elements of 3×4 matrix can be written as:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

Matrix notation

- Where there is no ambiguity a matrix can be represented by a single general element in brackets or by a capital letter in bold type.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

can be denoted by (a_{ij}) or by $\mathbf{A}$

Equal matrices

- Two matrices are equal if corresponding elements throughout are equal.

  $\mathbf{A} = \mathbf{B}$ that is $(a_{ij}) = (b_{ij})$ if $a_{ij} = b_{ij}$ for all values of i and j

Addition and subtraction of matrices

- Two matrices are added (or subtracted) by adding (or subtracting) corresponding elements. For example:

$$\begin{pmatrix} 4 & 2 & 3 \\ 5 & 7 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 8 & 9 \\ 3 & 5 & 4 \end{pmatrix} = \begin{pmatrix} 4+1 & 2+8 & 3+9 \\ 5+3 & 7+5 & 6+4 \end{pmatrix}$$
$$= \begin{pmatrix} 5 & 10 & 12 \\ 8 & 12 & 10 \end{pmatrix}$$

Multiplication of matrices

Scalar multiplication

- To multiply a matrix by a single number (a scalar), each individual element of the matrix is multiplied by that number. For example:

$$4\times\begin{pmatrix}3 & 2 & 5\\ 6 & 1 & 7\end{pmatrix}=\begin{pmatrix}12 & 8 & 20\\ 24 & 4 & 28\end{pmatrix}$$

That is:

$$k\left(a_{ij}\right)=\left(ka_{ij}\right)$$

Multiplication of matrices

Multiplication of two matrices

- Two matrices can only be multiplied when the number of columns in the first matrix equals the number of rows in the second matrix.
- The ijth element of the product matrix is obtained by multiplying each element in the ith row of the first matrix by the corresponding element in the jth column of the second matrix and the element products added.
- For example:

Multiplication of matrices

Multiplication of two matrices

$$\text{If } \mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} b_{11} \\ b_{21} \\ b_{31} \end{pmatrix}$$

$$\text{then } \mathbf{A.B} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} . \begin{pmatrix} b_{11} \\ b_{21} \\ b_{23} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} \end{pmatrix}$$

Multiplication of matrices

Multiplication of two matrices

$$\text{If } \mathbf{A} = \left(a_{ij}\right) \text{ is an } n \times m \text{ matrix and}$$
$$\mathbf{B} = \left(b_{ij}\right) \text{ is an } m \times q \text{ matrix then}$$
$$\mathbf{C} = \mathbf{A.B} = \left(c_{ij}\right) \text{ is an } n \times q \text{ matrix where}$$

$$c_{ij} = \sum_{k=1}^{m} a_{ik} b_{kj}$$

Transpose of a matrix

- If a new matrix is formed by interchanging rows and columns the new matrix is called the transpose of the original matrix. For example, if:

$$\mathbf{A} = \begin{pmatrix} 4 & 6 \\ 7 & 9 \\ 2 & 5 \end{pmatrix} \text{ then } \mathbf{A}^{\mathbf{T}} = \begin{pmatrix} 4 & 7 & 2 \\ 6 & 9 & 5 \end{pmatrix}$$

Special matrices

Square matrix

- A square matrix is of order m $\times$ m.
- A square matrix is symmetric if $a_{ij} = a_{ji}$. For example:

$$\begin{pmatrix} 1 & 2 & 5 \\ 2 & 8 & 9 \\ 5 & 9 & 4 \end{pmatrix}$$

- A square matrix is skew-symmetric if $a_{ij} = a_{ji}$. For example:

$$\begin{pmatrix} 0 & 2 & 5 \\ -2 & 0 & 9 \\ -5 & -9 & 0 \end{pmatrix}$$

Special matrices

Diagonal matrix

- A diagonal matrix is a square matrix with all elements zero except those on the leading diagonal. For example:

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 7 \end{pmatrix}$$

Special matrices

Unit matrix

- A unit matrix is a diagonal matrix with all elements on the leading diagonal being equal to unity. For example:

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- The product of matrix A and the unit matrix is the matrix A:

$$\mathbf{A.I} = \mathbf{A}$$

Special matrices

Null matrix

- A null matrix is one whose elements are all zero.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Notice that

$$\mathbf{A.0} = \mathbf{0}$$

- But that if $A.B = 0$ we cannot deduce that $A = 0$ or $B = 0$

Determinant of a square matrix

Singular matrix

- Every square matrix has its associated determinant. For example, the determinant of

$$\begin{pmatrix} 5 & 2 & 1 \\ 0 & 6 & 3 \\ 8 & 4 & 7 \end{pmatrix} \text{ is } \begin{vmatrix} 5 & 2 & 1 \\ 0 & 6 & 3 \\ 8 & 4 & 7 \end{vmatrix} = 150$$

- The determinant of a matrix is equal to the determinant of its transpose.
- A matrix whose determinant is zero is called a singular matrix.

Determinant of a square matrix

Cofactors

- Each element a_{ij} of a square matrix has a minor which is the value of the determinant obtained from the matrix after eliminating the ith row and jth column to which the element is common.
- The cofactor of element a_{ij} is then given as the minor of a_{ij} multiplied by

$$(-1)^{i+j}$$

Determinant of a square matrix

Adjoint of a square matrix

- Let square matrix C be constructed from the square matrix A where the elements of C are the respective cofactors of the elements of A so that if:

$$\mathbf{A}=\left(a_{ij}\right) \text{ and } A_{ij} \text{ is the cofactor of } a_{ij} \text{ then } \mathbf{C}=\left(A_{ij}\right)$$

- Then the transpose of C is called the adjoint of A, denoted by adjA.

Determinants

The symbol:

$$a_1b_2 - a_2b_1 = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \text{ (evaluated by cross multiplication as } \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \text{)}$$

- Is called a second-order determinant; second-order because it has two rows and two columns.

Determinants of third order

Minors

- A third-order determinant has three rows and three columns. Each element of the determinant has an associated *minor* – a second order determinant obtained by eliminating the row and column to which it is common. For example:

$$\text{the minor of } a_1 \text{ is } \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} \text{ obtained thus } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

Determinants of third order

Evaluation of a third-order determinant about the first row

- To expand a third-order determinant about the first row we multiply each element of the row by its minor and add and subtract the products as follows:

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

Determinants of third order

Evaluation of a determinant about any row or column

- To expand a determinant about any row or column we multiply each element of the row or column by its minor and add and subtract the products according to the pattern:

$$\begin{matrix} + & - & + & - & \cdots & \cdots & \cdots \\ - & + & - & + & \cdots & \cdots & \cdots \\ + & - & + & - & \cdots & \cdots & \cdots \\ - & + & - & + & \cdots & \cdots & \cdots \end{matrix}$$

Inverse of a square matrix

- If each element of the adjoint of a square matrix A is divided by the determinant of A then the resulting matrix is called the inverse of A, denoted by $A^{-}1$.

$$\mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}\left(\mathrm{adj}\mathbf{A}\right)$$

- Note: if det A = 0 then the inverse does not exist

Inverse of a square matrix

Product of a square matrix and its inverse

- The product of a square matrix and its inverse is the unit matrix:

$$\mathbf{A}.\mathbf{A}^{-1} = \mathbf{A}^{-1}.\mathbf{A} = \mathbf{I}$$

Solution of a set of linear equations

- The set of n simultaneous linear equations in n unknowns

$$\begin{array}{c} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = b_2 \\ \vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots + a_{nn}x_n = b_n \end{array}$$

- can be written in matrix form as:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \ldots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} \quad \text{that is } \mathbf{A.x} = \mathbf{b}$$

Solution of a set of linear equations

- Since:

$$\mathbf{A.x} = \mathbf{b} \quad \text{then}$$

$$\mathbf{A}^{-1}\mathbf{.Ax} = \mathbf{A}^{-1}\mathbf{.b} \quad \text{that is}$$

$$\mathbf{I.x} = \mathbf{A}^{-1}\mathbf{.b} \quad \text{and} \quad \mathbf{I.x} = \mathbf{x}$$

- The solution is then:

$$\boxed{\mathbf{x} = \mathbf{A}^{-1}\mathbf{.b}}$$

Solution of a set of linear equations

Gaussian elimination method for solving a set of linear equations

- Given:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

- Create the augmented matrix B, where:

$$\mathbf{B} = \left(\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} & b_n \end{array}\right)$$

Solution of a set of linear equations

Gaussian elimination method for solving a set of linear equations

$$\mathbf{B} = \left(\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} & b_n \end{array}\right)$$

- Eliminate the elements other than a11 from the first column by subtracting a_{21}/a_{11} times the first row from the second row, a_{31}/a_{11} times the first row from the third row, etc. This gives a new matrix of the form:

Solution of a set of linear equations

Gaussian elimination method for solving a set of linear equations

$$\left(\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ 0 & c_{22} & c_{23} & \cdots & c_{2n} & d_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & c_{n2} & c_{n3} & \cdots & c_{nn} & d_n \end{array}\right)$$

- This process is repeated to eliminate the c_{i2} from the third and subsequent rows until a matrix of the following form is arrived at:

$$\left(\begin{array}{ccccc|c} a_{11} & \cdots & a_{1,n-2} & a_{1,n-1} & a_{1n} & b_1 \\ 0 & \cdots & p_{n-3,n-2} & p_{n-2,n-1} & p_{n-2,n} & q_2 \\ 0 & \cdots & 0 & p_{n-1,n-1} & p_{n-1,n} & \vdots \\ 0 & \cdots & 0 & 0 & p_{nn} & q_n \end{array}\right)$$

Solution of a set of linear equations

Gaussian elimination method for solving a set of linear equations

$$\left(\begin{array}{ccccc|c} a_{11} & \cdots & a_{1,n-2} & a_{1,n-1} & a_{1n} & b_1 \\ 0 & \cdots & p_{n-3,n-2} & p_{n-2,n-1} & p_{n-2,n} & q_2 \\ 0 & \cdots & 0 & p_{n-1,n-1} & p_{n-1,n} & \vdots \\ 0 & \cdots & 0 & 0 & p_{nn} & q_n \end{array}\right)$$

- From this augmented matrix we revert to the product:

$$\begin{pmatrix} a_{11} & \cdots & a_{1,n-2} & a_{1,n-1} & a_{1n} \\ 0 & \cdots & p_{n-3,n-2} & p_{n-2,n-1} & p_{n-2,n} \\ 0 & \cdots & 0 & p_{n-1,n-1} & p_{n-1,n} \\ 0 & \cdots & 0 & 0 & p_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ q_2 \\ \vdots \\ q_n \end{pmatrix}$$

Solution of a set of linear equations

Gaussian elimination method for solving a set of linear equations

$$\begin{pmatrix} a_{11} & \cdots & a_{1,n-2} & a_{1,n-1} & a_{1n} \\ 0 & \cdots & p_{n-3,n-2} & p_{n-2,n-1} & p_{n-2,n} \\ 0 & \cdots & 0 & p_{n-1,n-1} & p_{n-1,n} \\ 0 & \cdots & 0 & 0 & p_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ q_2 \\ \vdots \\ q_n \end{pmatrix}$$

- From this product the solution is derived by working backwards from the bottom starting with:

$$p_{nn} x_n = q_n \quad \text{so} \quad x_n = \frac{q_n}{p_{nn}}$$

Eigenvalues and eigenvectors

- Expressed as a set of separate equations:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \lambda \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

- That is:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n &= \lambda x_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n &= \lambda x_2 \\ \vdots \qquad \vdots \qquad \vdots \qquad \vdots \qquad & \vdots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots + a_{nn}x_n &= \lambda x_n \end{aligned}$$

Eigenvalues and eigenvectors

- These can be rewritten as:

$$\begin{pmatrix} a_{11}-\lambda & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22}-\lambda & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn}-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

- That is:

$$(\mathbf{A}-\lambda\mathbf{I}).\mathbf{x}=\mathbf{0}$$

- Which means that, for non-trivial solutions:

$$|\mathbf{A}-\lambda\mathbf{I}|=0$$

Eigenvalues and eigenvectors

Eigenvalues

To find the eigenvalues of: $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

solve the characteristic equation: $\begin{vmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix} = 0$

That is: $(\lambda - 1)(\lambda - 5) = 0$

This gives eigenvalues $\boxed{\lambda_1 = 1;\ \lambda_2 = 5}$

To find the eigenvectors of $\mathbf{A}=\begin{pmatrix}4 & 1\\ 3 & 2\end{pmatrix}$ solve the equation $\mathbf{A.x}=\lambda\mathbf{x}$

For the eigenvalues $\lambda = 1$ and $\lambda = 5$

For λ=1

$$\begin{pmatrix}4 & 1\\ 3 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=1\begin{pmatrix}x_1\\ x_2\end{pmatrix} \text{ and so } x_2=-3x_1 \text{ giving eigenvector } \begin{pmatrix}k\\ -3k\end{pmatrix}$$

For λ=5

$$\begin{pmatrix}4 & 1\\ 3 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix}=5\begin{pmatrix}x_1\\ x_2\end{pmatrix} \text{ and so } x_2=x_1 \text{ giving eigenvector } \begin{pmatrix}k\\ k\end{pmatrix}$$

Diagonalization

- A matrix A is diagonalizable if and only if A has n linearly independent eigenvectors.
- In fact, $A = PDP^{-}1$ with D a diagonal matrix, if and only if the columns of P and n linearly independent eigenvectors of A. In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P.

Diagonalization

- **Example:** Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$$

- That is, find an invertible matrix P and a diagonal matrix D such that

$$A = PDP^{-1}$$

Solution

- Step 1. **Find the eigenvalues of A.**
- Here, the characteristic equation turns out to involve a cubic polynomial that can be factored:

Diagonalization

$$0 = \det(A - \lambda I) = -\lambda^3 - 3\lambda^2 + 4$$
$$= -(\lambda - 1)(\lambda + 2)^2$$

- The eigenvalues are $\lambda = 1$ and $\lambda = -2$.
- Step 2. **Find three linearly independent eigenvectors of A.**
- Three vectors are needed because A is a matrix.
- This is a critical step.
- If it fails, then Theorem 5 says that A cannot be diagonalized.

Diagonalization

- Basis for $\lambda = 1 : \mathrm{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

- Basis for $\lambda = -2 : \mathrm{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathrm{v}_3 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

- You can check that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set.

Diagonalization

- Step 3. **Construct P from the vectors in step 2.**
- The order of the vectors is unimportant.
- Using the order chosen in step 2, form

$$P = \begin{bmatrix} \mathrm{v}_1 & \mathrm{v}_2 & \mathrm{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

- Step 4. **Construct D from the corresponding eigenvalues.**
- In this step, it is essential that the order of the eigenvalues matches the order chosen for the columns of P.

Diagonalization

- Use the eigenvalue $\lambda = -2$ twice, once for each of the eigenvectors corresponding to $\lambda = -2$:

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

- To avoid computing P^{-1} , simply verify that $AD = PD$.
- Compute

$$AP = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ -1 & -2 & 0 \\ 1 & 0 & -2 \end{bmatrix} = PD$$

Thank You!